M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 2-

Day 3 report

Internship Program — Batch 2026

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1. Introduction

This report summarizes the activities completed during the **M-Tech Production & Marketing AI/ML Internship Program** on **08 July 2026**. During today's online session, I successfully studied and completed **Experiment 05** from the **Artificial Intelligence & Machine Learning Lab Manual**. The experiment focused on the concepts of **Supervised Learning**, including both **Regression** and **Classification** techniques. I gained theoretical knowledge and practical experience in building predictive models using popular Machine Learning algorithms such as **Linear Regression**, **Logistic Regression**, **Decision Trees**, **K-Nearest Neighbors (KNN)**, and **Support Vector Machines (SVM)**. I also learned how to evaluate model performance using metrics such as **R^2 Score**, **RMSE**, and **Accuracy**.

2. 2. Objectives

The primary objectives of today's learning session were:

- To understand the difference between **Regression** and **Classification** in Supervised Learning.
- To develop Regression models for predicting continuous numerical values.
- To develop Classification models for predicting categorical outcomes.
- To study the working principles of **Linear Regression**, **Logistic Regression**, **Decision Trees**, **K-Nearest Neighbors (KNN)**, and **Support Vector Machines (SVM)**.
- To understand the importance of dividing datasets into training and testing sets.
- To evaluate Machine Learning models using **R^2 Score**, **RMSE**, and **Accuracy**.
- To gain practical experience in implementing supervised learning algorithms using the **Scikit-learn** library.

3. 3. Task Description

Today's task involved studying **Experiment 05** from the AI/ML Lab Manual. The experiment introduced the concepts of **Supervised Learning**, with a primary focus on **Regression** and **Classification**.

I learned that **Regression** algorithms are used to predict continuous numerical values such as house prices, temperatures, and examination scores, whereas **Classification** algorithms predict discrete categories such as spam detection, disease diagnosis, and image classification.

The experiment also introduced five important supervised learning algorithms:

- Linear Regression
- Logistic Regression

- Decision Tree
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)

Additionally, I studied model evaluation techniques using **R^2 Score** and **RMSE** for regression problems, and **Accuracy** for classification models. I also learned the importance of splitting datasets into training and testing sets to evaluate model performance on unseen data.

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4. 4. Work Done

During today's internship session, I successfully completed **Experiment 05** from the AI/ML Lab Manual.

I began by studying the difference between **Regression** and **Classification**. I understood that Regression predicts continuous numerical values such as house prices, temperatures, and examination scores, whereas Classification predicts categorical outcomes such as spam detection, disease diagnosis, and image recognition.

Next, I explored **Linear Regression** in detail. I learned how a regression model fits a straight line through data to estimate continuous values. Using the **Scikit-learn** library, I practiced building a Linear Regression model, dividing the dataset into training and testing sets using `train_test_split`, and evaluating model performance using **R^2 Score** and **Root Mean Squared Error (RMSE)**. I understood that a higher R^2 Score indicates a better-fitting model, while a lower RMSE represents more accurate predictions.

I then studied several classification algorithms. I learned that **Logistic Regression** predicts class probabilities using a sigmoid (S-shaped) function, **Decision Trees** make predictions through a sequence of decision rules, **K-Nearest Neighbors (KNN)** classifies data based on the nearest neighboring points, and **Support Vector Machines (SVM)** identify the optimal boundary that maximizes the separation between different classes.

As part of the practical exercises, I trained a **KNN classifier** using the **Iris dataset** and successfully achieved perfect classification accuracy. I also implemented **Logistic Regression** and **Decision Tree** classifiers and compared their performance using **Accuracy Score**. These activities strengthened my understanding of dataset loading, model training, prediction, and performance evaluation using the **Scikit-learn** library.

Finally, I learned why Machine Learning models should always be tested on unseen data. This practice helps measure the model's ability to generalize to real-world situations while minimizing the risk of overfitting. I also explored the concept of **feature importance** in Decision Trees, which identifies the variables that contribute most significantly to the model's predictions.

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5. 5. Challenges Faced

The concepts presented in **Experiment 05** were well-organized and relatively easy to understand. However, understanding the mathematical intuition behind **Logistic Regression** and distinguishing between **R^2 Score** and **RMSE** required additional practice. I also spent extra time understanding how **K-Nearest Neighbors**

(KNN) calculates distances between data points and how selecting an appropriate value of **K** influences model accuracy.

Through repeated practice, reviewing examples, and experimenting with different datasets, I developed a much stronger understanding of these concepts. Implementing multiple supervised learning models and comparing their performance significantly enhanced my practical skills.

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6. 6. Key Learnings

During today's learning session, I achieved the following outcomes:

- Developed a clear understanding of the differences between **Regression** and **Classification**.
- Studied the working principles and applications of **Linear Regression**, **Logistic Regression**, **Decision Trees**, **K-Nearest Neighbors (KNN)**, and **Support Vector Machines (SVM)**.
- Learned how to split datasets into training and testing sets for reliable model evaluation.
- Gained practical experience in implementing supervised learning algorithms using **Scikit-learn**.
- Learned how to evaluate regression models using **R^2 Score** and **RMSE**.
- Learned how to evaluate classification models using **Accuracy**.
- Improved my understanding of model generalization, feature importance, and performance comparison.

These concepts form the essential foundation for implementing supervised Machine Learning solutions in real-world applications.

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7. 7. Conclusion

The successful completion of **Experiment 05** provided both theoretical knowledge and practical experience in **Supervised Machine Learning**. I now have a clear understanding of the differences between **Regression** and **Classification**, along with the ability to build, train, and evaluate models for both prediction tasks.

Working with **Linear Regression**, **Logistic Regression**, **Decision Trees**, **K-Nearest Neighbors (KNN)**, and **Support Vector Machines (SVM)** has strengthened my practical understanding of supervised learning techniques. The knowledge gained from this experiment has significantly increased my confidence in developing predictive Machine Learning models and applying them to real-world problems.

This learning provides a strong foundation for future experiments involving **Unsupervised Learning**, **Ensemble Learning**, and more advanced Artificial Intelligence applications throughout the internship program.

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9. Intern Information

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Date	**08 July, 2026**

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Domain	**AI/ML Intern**

Intern Signature ________ABD7________________